

NMEC595 - Research Methodology

Lecture 9: Probability

(Reference: Research Methodology, C.R. Kothari, 2nd Edition)

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Outline

- 1 Learning Objectives
- 2 Probability Notation and Events
- 3 Outcome Space and Set Operations
- 4 Interpretations of Probability
- 5 Basic Probability Rules
- 6 Conditional Probability
- 7 Independent Events

Learning Objectives

Upon completion of this lesson, you should be able to:

- 1 Translate events into probability notation and vice versa
- 2 Compute and interpret set operations (union, intersection, complement)
- 3 Determine if two events are independent, mutually exclusive, or neither
- 4 Compute and interpret conditional probabilities
- 5 Understand diagnostic testing metrics (sensitivity, specificity)
- 6

What is Probability?

Definition:

Probability is the likelihood of an outcome.

Key Concept:

- Probability quantifies uncertainty and helps us make decisions.
- Ranges from 0 (impossible) to 1 (certain).

Examples of Events

Definition:

An **event** is a collection of outcomes (denoted by capital letters A , B , C , etc.).

Example: Rolling Dice

Roll two fair six-sided dice.

Define event $R = \{\text{rolling doubles}\}$

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

Converting to Probability Notation

- 1 **Identify** the outcome event of interest
 - Example: “Rolling a double with two dice”
- 2 **Use a letter** to represent this outcome
 - Example: $R = \{\text{Rolling a double}\}$
- 3 **State the probability** using notation
 - Example: $P(R) = \text{“Probability of rolling a double with 2 dice”}$

Outcome Space (Sample Space)

definition The **outcome space** S (or **sample space**) is the set of all possible outcomes that can occur in an experiment.

Example: Rolling Two Dice

Each die shows 1–6, so ordered pairs (i, j) where $i, j \in \{1, 2, 3, 4, 5, 6\}$

$$S = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$$

Total outcomes: $|S| = 6 \times 6 = 36$

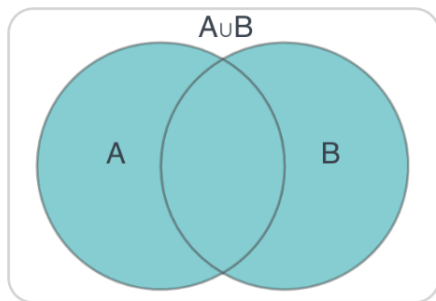
Set Operation: Union

Definition

The **union** of events A and B contains all outcomes in A , B , or **both**.

Notation: $A \cup B$ (read as “ A or B ”)

Key: In statistics, “or” means **at least one** occurs (inclusive or)

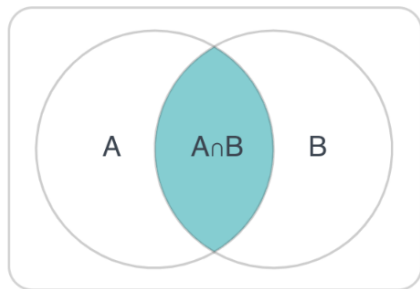


Set Operation: Intersection

Definition

The **intersection** of events A and B contains all outcomes that are in both A and B .

Notation: $A \cap B$ (read as “ A and B ”)

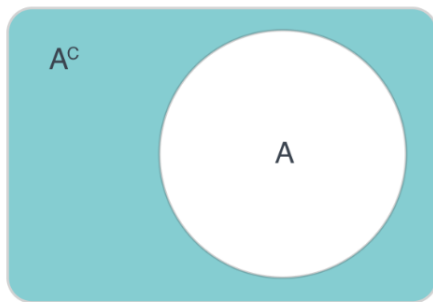


Set Operation: Complement

Definition

The **complement** of event A contains all outcomes that are **not** in A .

Notation: A^c , $\bar{A}$, or A'

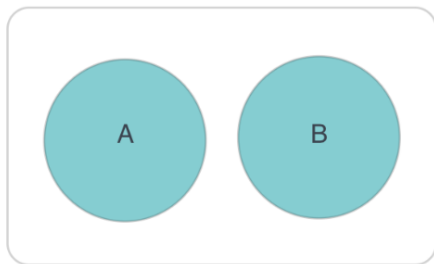


Set Operation: Mutually Exclusive

Definition

Events A and B are **mutually exclusive** if they have no outcomes in common. They cannot both occur simultaneously.

Notation: $A \cap B = \emptyset$ (empty set)



Example: Two Dice Events

Rolling two fair six-sided dice, define:

$$A = \{(3, 5)\}$$

$$B = \{4 \text{ on first die}\} = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\}$$

$$C = \{5 \text{ on second die}\} = \{(1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (6, 5)\}$$

$$D = \{\text{sum} = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

Find: $B \cap D$ and $B \cup D$

Solution:

- $B \cap D = \{(4, 3)\}$ (only outcome in both)
- $B \cup D =$
 $\{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (1, 6), (2, 5), (3, 4), (5, 2), (6, 1)\}$
 (11 outcomes)

Three Interpretations of Probability

1. Classical (Theoretical)

When all outcomes are equally likely:

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of possible outcomes}}$$

Example: Fair coin, fair dice, drawing cards

2. Relative Frequency (Empirical)

Based on observed data over many trials:

$$P(E) \approx \frac{\text{number of times } E \text{ occurred}}{\text{total number of trials}}$$

Example: Flip coin 10,000 times: 4,555 heads $\Rightarrow P(H) \approx 0.4555$

Three Interpretations (continued)

3. Subjective Probability

Based on personal belief, judgment, information, intuition, and experience.

Examples of Subjective Probability

- What is $P(\text{you get an A in this course})$?
- What is $P(\text{it will rain tomorrow})$?
- What is $P(\text{your startup succeeds})$?

Each person may assign different probabilities based on their knowledge and beliefs.

Note

All three interpretations follow the same mathematical rules of probability!

Exercise: Classical Probability

Problem

Find the probability that the sum of two fair dice is greater than or equal to 10.

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Problem

Find the probability that the sum of two fair dice is greater than or equal to 10.

Solution:

- Total outcomes: 36
- $P(S = 10) = P\{(4, 6), (5, 5), (6, 4)\} = \frac{3}{36}$
- $P(S = 11) = P\{(5, 6), (6, 5)\} = \frac{2}{36}$
- $P(S = 12) = P\{(6, 6)\} = \frac{1}{36}$

$$P(S \geq 10) = \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{6}{36} = \frac{1}{6} \approx 0.167$$

Fundamental Probability Properties

Property 1: Range

For any event A :

$$0 \leq P(A) \leq 1$$

- $P(A) = 0$: Event is impossible
- $P(A) = 1$: Event is certain

Property 2: Complement Rule

$$P(A) = 1 - P(A^c)$$

Equivalently: $P(A) + P(A^c) = 1$

Union Rule and Special Cases

Property 3: General Union Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Why subtract? To avoid double-counting the intersection.

Special Case: Mutually Exclusive Events

If A and B are mutually exclusive (i.e., $A \cap B = \emptyset$), then:

$$P(A \cup B) = P(A) + P(B)$$

because $P(A \cap B) = 0$

Practice Problem

Given: $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$

Find:

- ① $P(A^c)$
- ② $P(A \cup B)$
- ③ $P(A \cap B^c)$
- ④ $P(B \cap A^c)$

Practice Problem

Given: $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$

Find:

- ① $P(A^c)$
- ② $P(A \cup B)$
- ③ $P(A \cap B^c)$
- ④ $P(B \cap A^c)$

Solutions:

- ① $P(A^c) = 1 - P(A) = 1 - 0.6 = 0.4$
- ② $P(A \cup B) = 0.6 + 0.5 - 0.2 = 0.9$
- ③ $P(A \cap B^c) = P(A) - P(A \cap B) = 0.6 - 0.2 = 0.4$
- ④ $P(B \cap A^c) = P(B) - P(A \cap B) = 0.5 - 0.2 = 0.3$

What is Conditional Probability?

Definition (Conditional Probability)

The **conditional probability** of event A given that event B has occurred.

Notation: $P(A|B)$ (read as “probability of A given B ”)

Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Intuition: We restrict our attention to outcomes where B occurred, then ask how often A also occurs.

Real-World Conditional Probability

Example 1: Traffic

Question: What's the quickest route to work?

- If it's rush hour $\Rightarrow$ Route changes (conditional)
- If it's 3 AM $\Rightarrow$ Same route works (less conditional dependence)

Example 2: Sports

Question: What's the probability a team wins the championship?

- Outdoor stadium + rain forecast $\Rightarrow$ Probability changes (conditional)
- Indoor stadium $\Rightarrow$ Weather doesn't matter (independent)

Example: World Campus Enrollment

Fall 2024 Enrollment Data:

	Female	Male	Total
Undergraduate	3814	3428	7242
Graduate	2213	2787	5000
Total	6027	6215	12242

Questions:

- 1 What is $P(F)$? (marginal probability)
- 2 What is $P(F|G)$? (conditional probability)

Example: Enrollment (continued)

Solution:

1. Marginal probability:

$$P(F) = \frac{6027}{12242} = 0.492$$

2. Conditional probability:

$$P(F|G) = \frac{\text{Female graduates}}{\text{Total graduates}} = \frac{2213}{5000} = 0.443$$

Important Observation

$$P(F) = 0.492 \neq 0.443 = P(F|G)$$

This means gender and level are **dependent** (not independent)!

Example: Rolling Two Dice

Define events:

- $C = \{5 \text{ on second die}\}$
- $D = \{\text{sum} = 7\}$

Find: $P(C|D)$

Example: Rolling Two Dice

Define events:

- $C = \{5 \text{ on second die}\}$
- $D = \{\text{sum} = 7\}$

Find: $P(C|D)$

Solution (Approach 1 - Using sample space):

- $D = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ (6 outcomes)
- Given D occurred, only $(2, 5)$ has 5 on second die
- $P(C|D) = \frac{1}{6}$

Solution (Approach 2 - Using formula):

- $P(D) = \frac{6}{36}$, $P(C \cap D) = \frac{1}{36}$ (only $(2, 5)$)
- $P(C|D) = \frac{P(C \cap D)}{P(D)} = \frac{1/36}{6/36} = \frac{1}{6} \checkmark$

Definition of Independence

Definition (Independent Events)

Events A and B are **independent** if the occurrence of one does not affect the probability of the other.

To verify independence, show ANY ONE of:

- ① $P(A \cap B) = P(A) \cdot P(B)$
- ② $P(A|B) = P(A)$
- ③ $P(B|A) = P(B)$

If ANY ONE fails, events are dependent!

Example: Independent Events (Coin Flips)

Problem

Find $P(\text{two tails in two coin flips})$

Solution:

- Let $A = \{\text{tail on flip 1}\}$, $P(A) = \frac{1}{2}$
- Let $B = \{\text{tail on flip 2}\}$, $P(B) = \frac{1}{2}$
- Coin flips are **independent** (flip 1 doesn't affect flip 2)

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Check: From sample space $\{HH, HT, TH, TT\}$, only TT gives two tails:
 $\frac{1}{4}$ ✓

Example: Dependent Events (College Admissions)

Problem

You apply to Harvard (H) and Penn State (P).

- $P(H) = 0.3$ (admitted to Harvard)
- $P(P) = 0.6$ (admitted to Penn State)
- $P(H \cap P) = 0.25$ (admitted to both)

Are these events independent?

Solution:

Check if $P(H \cap P) = P(H) \cdot P(P)$:

$$P(H) \cdot P(P) = 0.3 \times 0.6 = 0.18$$

But $P(H \cap P) = 0.25 \neq 0.18$

⇒ Events are **DEPENDENT**!

Independent vs. Mutually Exclusive

Critical Distinction

Independent and **mutually exclusive** are **DIFFERENT** concepts!

Mutually Exclusive:

- Cannot occur together
- $A \cap B = \emptyset$
- $P(A \cap B) = 0$
- Events are **dependent**

Independent:

- One doesn't affect other
- Can occur together
- $P(A \cap B) = P(A) \cdot P(B)$
- $P(A|B) = P(A)$

Example: Drawing Without Replacement

Problem

An urn contains 6 red balls and 4 blue balls. Two balls are picked randomly **without replacement**. Find $P(\text{both red})$.

Solution:

- Let $R_1 = \{\text{red on first draw}\}$
- Let $R_2 = \{\text{red on second draw}\}$
- Events are **dependent** (without replacement!)

$$\begin{aligned}P(R_1 \cap R_2) &= P(R_1) \cdot P(R_2|R_1) \\&= \frac{6}{10} \times \frac{5}{9} \\&= \frac{30}{90} = \frac{1}{3} \approx 0.333\end{aligned}$$